

Himesh Ahuja

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Education

University of Virginia <i>Computer Science and Mathematics / GPA: 3.82 / 4.0</i> Honors: Echols Scholar (Top 5% of Class), College Science Scholar (~12 Students), Ingrassia Research Grant (\$2K Award) Coursework: Data Structures & Algorithms, Computer Architecture, Software Development, Intermediate Economics Involvement: Virginia Poker Club, Climbing Club, Intramural Basketball, Indian Student Association	August 2025 – May 2028 <i>Charlottesville, VA</i>
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Experience

AI/ML Software Engineer Intern <i>Noblis</i> - Built a secure Python sandbox API in FastAPI handling 250K+ requests, enabling safe LLM agent code execution with full process isolation - Containerized sandbox environments and web apps using Docker and systemd; wrote 30+ PyTest tests to validate execution safety and prevent regressions - Developed a Playwright-based LLM evaluation framework and automated runs via GitHub Actions; engineered TypeScript middleware for vLLM servers on 10+ B200 GPUs, leading to 25% quicker deployments and QA	January 2026 – Present <i>Reston, VA</i>
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Software Developer <i>theCourseForum</i> - Engineer backend features for UVA’s largest course analytics platform, serving 20K+ students (85% of the student body) - Optimized AWS ECS deployment pipeline by caching dependencies and implementing multistage Docker builds, cutting deployment time by 45% and eliminating disk-space build failures entirely - Built a reply system supporting 100K+ course reviews, enabling scalable, user discussions across the course catalog	September 2025 – Present <i>Charlottesville, VA</i>
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Research Assistant <i>Chandra Robot Autonomy Lab</i> - Deployed full ROS2 autonomy stack on Clearpath Jackals and Traxxas vehicles in Python; awarded \$2K Ingrassia Research Grant for contributions to high-speed autonomous navigation - Processed LiDAR 3D point clouds into 2D occupancy maps via SLAM and RViz, enabling real-time obstacle avoidance validated across 50+ autonomous test runs - Tuned Nav2 path planners to raise success rates by ~25% on Jackals, unlocking reliable navigation at 50% higher speeds	August 2025 – Present <i>Charlottesville, VA</i>
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Software Engineer Intern <i>NT Concepts</i> Delivered 10+ GIS features on a Google subcontracting project using JavaScript, React.js, and Electron; demoed to 100+ internal staff; led testing and debugging across 15+ components with Jest, cutting QA cycle time by ~30%	May 2024 – August 2024 <i>Tysons Corner, VA</i>
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Projects

A8THER <i>Python, FastAPI, Docker, Ghidra, LangChain, WebSockets, Xterm.js, SQLite</i> - Built an agentic AI reverse engineering tool that decompiles unknown binaries into readable code and natural-language explanations — reducing expert analysis time from hours to seconds - Orchestrated an LLM reasoning loop over Ghidra, objdump, and readelf inside a Docker sandbox; cached results in SQLite with dynamic ARM64/AMD64 selection, cutting repeat latency by ~60%	HooHacks 2026
Energy Analytics Platform <i>scikit-learn, Streamlit, Pandas, Matplotlib</i> Applied Matrix Profile, LOF, and K-means clustering to 200K+ rows of time-series energy data to surface anomalies and long-term consumption trends, then built a Streamlit dashboard for real-time anomaly detection and trend visualizations	Machine Learning @ UVA
CorporateHistory RAG Engine <i>LangChain, Supabase PostgreSQL, Next.js, Python</i> Built a pro-bono RAG system over 10,000+ unstructured records using LangChain and Supabase pgvector, delivering low-latency semantic search through a Next.js UI	Business and AI Institute

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, Dart, HTML, CSS, x86-64 Assembly, ROS2
Frameworks & Libraries: React.js Node.js, FastAPI, Flask, Django, LangChain, TensorFlow, PyTorch, OpenCV, Pandas, NumPy, Scikit-learn, Playwright,
Databases & Infrastructure: PostgreSQL, Supabase, MySQL, SQLite, Firebase, MongoDB, AWS (ECS, S3), Docker, Kubernetes/K8s, GitHub Actions, Ghidra, Foxglove, Git, Vim, tmux, lldb/gdb
Concepts: Agentic AI, RAG Systems, REST & GraphQL APIs, Machine Learning, LLM Evaluation, CI/CD, Cloud Computing, Sandboxed Execution, SLAM, OOP, Agile/Scrum